

Superantigens and Immune Evasion in *S. aureus*

Although conventional antigens require recognition by all five elements of the T-cell-receptor complex, superantigens bind directly to the variable region of the β chain of the T-cell receptor. This unique interaction bypasses normal antigen processing and presentation, leading to the activation of 5% to 30% of resting T cells. In contrast, a typical antigenic response activates only 0.0001% to 0.01% of T cells.

Non-specific T-cell activation results in a massive systemic release of cytokines, including: - **Interleukin-2 (IL-2)** - **Interferon- γ (IFN- γ)** - **Tumor necrosis factor- β (TNF- β)** from T cells

- **Interleukin-1 (IL-1)** and **tumor necrosis factor- α (TNF- α)** from macrophages

Superantigens also drive the expansion of lymphocytes expressing specific T-cell-receptor variable regions of the β chain. They can activate B cells, potentially leading to elevated levels of **immunoglobulin E (IgE)** or production of **autoantibodies**. Additionally, superantigens have been shown to selectively induce **cutaneous lymphocyte-associated antigen (CLA)** on T cells, directing their migration ("homing") to the skin.

Immune Evasion Mechanisms of *S. aureus*

S. aureus employs multiple strategies to evade immune clearance: - Approximately 60% of strains secrete **chemotaxis inhibitory protein of staphylococci (CHIPS)**, which blocks neutrophil chemotaxis. - **Protein A** binds the Fc region of IgG, preventing opsonization. - **Staphylokinase** promotes fibrinolysis and interferes with phagocytosis. - **Capsular polysaccharide, fibrinogen-binding protein, and clumping factor A** help resist opsonophagocytosis. - **Staphylokinase** and **aureolysin** bind and cleave antimicrobial peptides, enhancing bacterial survival both in vitro and likely in vivo.

Drug Resistance in *S. aureus*

Antibiotic resistance is a major challenge in managing staphylococcal infections. The widespread use of penicillins has led to the emergence of **methicillin-resistant *S. aureus* (MRSA)**, a significant epidemiological concern since the 1980s. Methicillin resistance confers **pan-resistance to all β -lactam antibiotics**.

More recently, **intermediate-level resistance to vancomycin (VISA)** has emerged, posing an additional therapeutic threat. While overall *S. aureus* infection rates remain stable, the proportion of MRSA isolates has increased in several countries.

High-risk groups for MRSA colonization or infection include: - Adults over 65 years - Chronically ill individuals - Patients with prolonged hospital stays (especially post-surgery) - Exposure to MRSA-positive individuals - Prior antibiotic use

The primary reservoirs for MRSA are **healthcare workers** and **hospitalized patients**, with transmission occurring mainly via direct contact.

Efforts to eradicate MRSA have largely failed. Topical **mupirocin** applied to the anterior nares and wounds reduces *S. aureus* colonization, but studies show it may not prevent transmission—for example, in long-term care facilities. Although mupirocin is effective in decolonization, its indiscriminate use should be avoided due to the rising incidence of **mupirocin-resistant strains**.

Immunity Against *S. aureus*

The primary defense against *S. aureus* is **innate immunity**, particularly mediated by **neutrophils**. A key component of this response is **antimicrobial peptides**, such as: - Dermicidin - LL-37 - Protegrin - α -defensins and β -defensins - Lactoferricin - Cascocidin

Despite the presence of circulating antibodies to staphylococcal cell-wall antigens and extracellular toxins in most adults—indicating frequent exposure—these antibodies are **not sufficient for protective immunity**.

Emerging evidence shows that *S. aureus* can **invade and survive within host cells**, including epithelial and immune cells. This intracellular persistence suggests that **cell-mediated immunity** is essential for clearing the infection.

Clinical Findings

Superficial Staphylococcal Pyodermas

Pyodermas are infections localized to the epidermis, just beneath the stratum corneum, or within hair follicles.